

CATHERINE WOLFRAM

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Last updated: June 4, 2026

EMPLOYMENT

ETH Zurich , Zurich, Switzerland Postdoc	July 2025-July 2026
Yale University , New Haven, CT NSF Postdoctoral Fellow and Gibbs Assistant Professor	July 2024 - Present (on leave July 2025-July 2026)

EDUCATION

Massachusetts Institute of Technology , Cambridge, MA PhD in Mathematics Advisor: Scott Sheffield Thesis title: <i>Random geometry in two and three dimensions</i>	August 2019 - May 2024
University of Chicago , Chicago, IL B.A. in Mathematics with Honors Dean's list, 2015-2019	September 2015 - June 2019
Stanford University Online High School	September 2011 - June 2015

ARTICLES

- The multinomial dimer model**, with Richard Kenyon. ArXiv preprint 2506.12171.
- Epstein curves and holography of the Schwarzian action**, with Franco Vargas Pallete and Yilin Wang. ArXiv preprint 2503.05690.
- Large deviations for the 3D dimer model**, with Nishant Chandgotia and Scott Sheffield. *Accepted* (to appear) in **Asterisque**.
- Circle homeomorphisms with square summable diamond shears**, with Dragomir Šarić and Yilin Wang. **Int. Math. Res. Not. (IMRN)** Vol. 2024, 17 (2024).

AWARDS

NSF Postdoctoral Fellowship	Fall 2024-Spring 2028
Baddoo Community Buidling Award MIT math department award	May 2024
Phyllis Ruby Block Fellowship MIT math department fellowship	Spring 2024

TEACHING

Yale:

- Instructor, Discrete mathematics (Math 244) Fall 2026
- Instructor, Vector analysis and integration on manifolds (Math 302), Spring 2027

MIT:

- Recitation instructor, probability and random variables (18.600) Fall 2023
Instructor: Scott Sheffield
- Recitation instructor, multivariable calculus (18.02A) January 2023
Instructor: Bill Minicozzi
- Recitation instructor, probability and random variables (18.600) Spring 2022
Instructor: Scott Sheffield
- Recitation instructor, probability and random variables (18.600) Fall 2021
Instructor: Jonathan Kelner

University of Chicago:

- Grader for Honors Calculus (Math 162) January 2019 - March 2019
- Course assistant (VCA) for Calculus (Math 151-152-153) September 2016 - June 2017

MENTORSHIP AND PROFESSIONAL EXPERIENCE

Referee for journal articles Periodic
Annals of Probability, Transactions of the American Mathematical Society, Discrete Mathematics, Proceedings of the London Mathematical Society

Defense committee member for graduate students
Raphael de Marreiros (PhD February 2025)

Mentor for students
Sumukh Ramakrishnan and Jiakai Li (ETH masters students, semester project, Spring 2026), June Lee (Yale undergraduate, reading project, summer 2025), MIT undergraduates through the directed reading program (2020-2024).

Mentor for Research Science Institute (RSI) hosted by MIT (summer 2020)

SERVICE AND ORGANIZATION

Organizer of informal seminars at Yale and ETH 2025-Present
Started and ran an informal seminar at Yale (Spring 2025) and then at ETH (Fall 2025-Spring 2026), which meet every week, for PhD students and postdocs to present on papers or their work in a range of topics (random conformal geometry, statistical mechanics, hyperbolic geometry, geometric analysis, etc...)

Co-organizer of the Friday Morning Seminar at Yale September 2024 - May 2025
Informal weekly seminar with some invited speakers and some informal discussions on topics in geometry, probability, combinatorics, dynamics, etc.

Co-organizer and mentor, MIT Directed Reading Program (DRP) October 2022 - May 2024
Evaluated applications, paired undergraduate students with mentors, and arranged events. The program has about 100 student per year.

Co-organizer of PuMaGraSS at MIT September 2020 - May 2021
Weekly, casual seminar for MIT math graduate students to give talks to their peers, name stands for "Pure Math Graduate Student Seminar."

Manager of the Math Learning Center (MLC) at MIT August 2020 - May 2021
Ran the math department's peer-tutoring center, managing 10-12 undergraduate tutors.

TALKS

Past and upcoming talks as of now, listed in reverse chronological order.

1. Cambridge equinox probability conference, Cambridge, UK	September 2026
2. Random Explorations: From Random Walks to Random Geometry, IMSI, Chicago, IL	July 2026
3. One World Probability Seminar (online talk)	April 2026
4. Random Geometry in Math and Physics, Simons Center, Stony Brook, NY	April 2026
5. EPFL probability seminar, Lausanne, Switzerland	March 2026
6. University of Bern IMSV colloquium, Bern, Switzerland	March 2026
7. Les Diablerets workshop on mathematical physics, Les Diablerets, Switzerland	February 2026
8. IPAM workshop: New interactions between probability and geometry, Los Angeles, CA	January 2026
9. KTH probability and mathematical physics seminar, Stockholm, Sweden	December 2025
10. Vienna probability seminar, Vienna, Austria	November 2025
11. Universal Teichmüller theory workshop, Aussois, France	August 2025
12. CIRM perfectly matched perspectives in statistical mechanics, combinatorics and geometry, Marseille, France	June 2025
13. AMS sectional meeting session on Probability and Combinatorics Hartford, CT	April 2025
14. University of Michigan probability seminar, Ann Arbor, MI	February 2025
15. Princeton probability seminar, Princeton, NJ	November 2024
16. Columbia probability seminar, New York, NY	October 2024
17. AdS geometry reading group (two expository talks, online)	October 2024
18. Northwestern probability seminar, Evanston, IL	September 2024
19. 11th World Congress on Probability and Statistics, Bochum, Germany	August 2024
20. Workshop on Probabilistic Field Theories, Aalto University	June 2024
21. Statistical physics and random surfaces, Oberwolfach	May 2024
22. IPAM workshop: Statistical Mechanics Beyond 2D, Los Angeles, CA	May 2024
23. Seed seminar of Mathematics and Physics (online)	April 2024
24. PuMaGraSS seminar at MIT (expository talk) <i>SLE curves, large deviations, and geometry</i>	March 2024
25. KPZ meets KPZ workshop, Fields Institute, Toronto, Canada	March 2024
26. University of Edinburgh probability seminar (online)	January 2024
27. MIT probability seminar, Cambridge, MA	November 2023
28. University of Chicago probability and statistical physics seminar Chicago, IL	November 2023
29. Brown combinatorics seminar, Providence, RI	November 2023
30. Informal seminar: dynamics, geometry, and moduli spaces at Harvard	November 2023

31. ICERM workshop: Asymptotic Limits of Discrete Random Structures Providence, RI	September 2023
32. Harvard CMSA probability seminar, Cambridge, MA	September 2023
33. MIT integrable probability seminar, Cambridge, MA	September 2023
34. dimers2023 conference, Sorbonne Universite, Paris, France	July 2023
35. Percolation today (online)	June 2023
36. LA probability forum (joint between Caltech, UCLA, and USC)	May 2023
37. IHES informal analysis and probability seminar, Bures sur Yvette, France	April 2023
38. Courant probability and mathematical physics seminar, New York, NY	March 2023
39. Rutgers geometry/topology seminar (online)	March 2023
40. Informal seminar: dynamics, geometry, and moduli spaces at Harvard	February 2023
41. CUNY Hyperbolic geometry seminar, New York, NY	February 2023
42. Random Geometry and Statistical Physics Workshop, University of Pennsylvania	October 2022
43. Probability and Mathematical Physics conference in Helsinki, Finland	July 2022
44. PuMaGraSS seminar at MIT (expository talk) <i>Gluing polygons, Wick's formula, and moduli spaces</i>	April 2022
45. Reading group on the dimer model at MIT (expository talk) <i>Kasteleyn theory on the torus</i>	March 2022
46. PuMaGraSS seminar at MIT (expository talk) <i>The dimer model in 2D and 3D, local moves, and topology</i>	April 2021
47. AWM Women in Math Symposium, University of Chicago <i>Recursive derivation of $2n$-gon topologies.</i>	February 2017

ACADEMIC VISITS

Institute for Advanced Study, Princeton, NJ September 2022 - December 2022
Visiting graduate student while my advisor was there as a visiting professor.

Institut des Hautes Études Scientifiques, Bures sur Yvette, France April - May 2023; June 2024
Invited researcher visiting Prof. Yilin Wang.

CONFERENCES AND SUMMER SCHOOLS

A more complete list of conferences, summer schools, and workshops I've attended or plan to attend as of now, in reverse chronological order.

Cambridge equinox probability conference, Cambridge, UK	September 2026
Random Explorations: From Random Walks to Random Geometry, IMSI, Chicago, IL	July 2026
Random Geometry in Math and Physics, Simons Center, Stony Brook, NY	April 2026
Les Diablerets workshop on mathematical physics, Les Diablerets, Switzerland	February 2026
IPAM workshop: New interactions between probability and geometry, Los Angeles, CA	January 2026
Universal Teichmüller theory workshop, Aussois, France	August 2025

CIRM perfectly matched perspectives on statistical mechanics, combinatorics and geometry	June 2025
AMS sectional meeting session on Probability and Combinatorics	April 2025
Complex Faces, Venice, Italy	April 2025
Northeast Probability Seminar, New York, NY	November 2024
11th World Congress on Probability and Statistics, Bochum, Germany	August 2024
Workshop on Probabilistic Field Theories, Aalto University, Espoo, Finland	June 2024
Statistical physics and random surfaces, Oberwolfach, Germany	May 2024
Statistical mechanics beyond 2D, IPAM, Los Angeles, CA	May 2024
KPZ meets KPZ workshop, Fields Institute, Toronto, Canada	March 2024
Probability and geometry in, on and of non-Euclidean spaces, CIRM, Marseille, France	October 2023
Asymptotic limits of discrete random structures, ICERM, Providence, RI	September 2023
dimers2023 conference, Sorbonne Universite, Paris, France	July 2023
Three facets of gravity workshop, Humboldt University, Berlin, Germany	May 2023
Random Geometry and Statistical Physics Workshop, University of Pennsylvania, Philadelphia, PA	October 2022
Cornell Probability Summer School (as a TA), Cornell University, Ithaca, NY	August 2022
Probability and Mathematical Physics conference, Helsinki, Finland	July 2022
The Analysis and Geometry of Random Spaces, MSRI, Berkeley, CA	March 2022
Program associate for two weeks of the semester program (including the one week workshop)	
Random Geometry conference, CIRM, Marseille, France	January 2022
Random Conformal Geometry Summer School, MSRI, Berkeley, CA (virtual)	July 2021

OTHER

Citizenships: United States, United Kingdom